

THREE RIVERS, MI, USA

WMO: 720371

Lat: 41.960N

Lon: 85.593W

Elev: 251

StdP: 98.34

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 54850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-15.0	-22.2	0.5	-17.4	-18.9	0.7	-14.2	10.9	-2.6	9.3	-3.3	2.2	210	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.0	22.5	30.5	21.8	29.0	21.1	24.4	29.3	23.6	28.1	22.8	26.9	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	26.6	22.3	17.5	26.0	21.4	16.5	25.1	74.9	29.3	71.5	28.1	68.5	27.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.1	-23.0	34.1	3.1	1.9	-25.3	35.5	-27.1	36.6	-28.9	37.7	-31.1	39.1		
(5)			-23.3	25.8	3.1	1.0	-25.5	26.6	-27.3	27.2	-29.0	27.7	-31.2	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-4.3	-3.4	3.0	9.1	15.9	20.6	22.3	21.4	17.9	11.4	4.3	-0.5	(6)
(7)		DBStd	10.59	6.25	6.18	6.21	5.02	4.87	3.35	3.21	2.90	4.17	4.86	4.90	5.05	(7)
(8)		HDD10.0	1681	442	375	231	76	8	0	0	0	1	42	181	327	(8)
(9)		HDD18.3	3481	700	607	475	280	107	16	5	8	58	222	420	584	(9)
(10)		CDD10.0	1641	1	1	16	48	190	318	382	352	238	84	11	1	(10)
(11)		CDD18.3	399	0	0	1	2	31	84	128	102	46	6	0	0	(11)
(12)		CDH23.3	3462	0	0	11	25	324	686	1182	796	384	54	1	0	(12)
(13)		CDH26.7	1008	0	0	1	2	83	196	402	205	112	8	0	0	(13)
(14)	Wind	WSAvg	2.8	3.4	3.4	3.3	3.7	2.9	2.5	1.9	1.8	2.0	2.7	3.0	3.2	(14)
(15)	Precipitation	PrecAvg	1001	69	54	64	82	115	100	101	106	89	87	76	63	(15)
(16)		PrecMax	1201	144	116	121	148	219	193	226	324	296	188	144	119	(16)
(17)		PrecMin	818	25	21	20	10	41	44	39	19	27	18	27	19	(17)
(18)		PrecStd	120	35	29	25	33	48	43	46	60	59	46	32	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.0	24.8	26.3	31.3	32.5	35.3	32.4	32.4	28.0	21.1	15.4	(19)
(20)			MCWB	11.7	12.1	18.0	16.0	19.8	21.9	23.0	22.5	22.3	19.7	14.4	13.4	(20)
(21)		2%	DB	9.5	11.9	19.8	22.9	28.8	30.9	32.3	30.5	30.1	25.0	17.5	11.8	(21)
(22)			MCWB	8.1	10.0	14.6	14.7	19.0	21.1	23.4	22.1	22.2	17.7	13.0	9.8	(22)
(23)		5%	DB	6.2	7.6	15.8	20.9	27.2	28.9	30.7	28.9	27.5	22.3	14.7	8.7	(23)
(24)			MCWB	4.6	5.7	11.5	13.6	18.5	20.8	22.4	21.6	20.4	16.9	11.0	7.2	(24)
(25)		10%	DB	3.8	4.6	12.5	17.8	24.9	27.3	28.8	27.5	25.4	19.8	12.0	6.2	(25)
(26)			MCWB	2.6	2.8	9.0	12.0	17.6	19.9	21.4	20.7	19.1	15.7	9.2	4.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.9	18.4	18.4	22.2	24.5	25.8	25.0	24.2	21.7	16.4	14.1	(27)
(28)			MCD B	13.1	14.2	24.6	23.3	27.3	29.8	31.1	29.4	29.9	26.3	18.3	15.4	(28)
(29)		2%	WB	8.5	9.4	15.5	16.4	21.0	23.4	24.8	24.0	23.1	19.8	13.6	10.0	(29)
(30)			MCD B	9.1	11.8	18.7	20.9	26.0	28.0	29.8	28.0	27.9	22.5	16.8	11.2	(30)
(31)		5%	WB	5.1	5.8	12.6	14.7	19.9	22.4	23.8	23.1	21.9	17.9	11.4	7.1	(31)
(32)			MCD B	6.2	7.4	15.0	19.0	24.7	26.7	28.6	26.8	25.5	20.4	13.9	8.2	(32)
(33)		10%	WB	2.7	2.9	9.3	13.0	18.7	21.4	22.8	22.2	20.8	16.1	9.1	4.8	(33)
(34)			MCD B	3.7	4.4	12.3	17.7	23.4	25.2	27.1	25.7	24.0	19.5	11.8	6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	9.4	11.0	11.4	10.4	11.1	10.9	11.4	10.1	8.2	6.4	(35)
(36)			MCDBR	8.6	10.9	13.8	14.8	14.1	12.1	12.4	12.2	13.0	13.3	12.2	9.8	(36)
(37)			MCWBR	7.5	9.0	9.1	8.3	6.5	5.1	4.8	5.3	5.9	7.3	8.4	8.4	(37)
(38)		5% WB	MCDBR	8.3	10.7	12.2	12.7	11.8	9.8	10.5	9.4	10.4	10.8	11.2	9.1	(38)
(39)			MCWBR	7.7	9.2	9.0	9.0	6.6	5.2	5.0	4.9	5.5	7.5	8.6	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.346	0.352	0.393	0.426	0.439	0.435	0.441	0.400	0.354	0.319	0.301	(40)	
(41)		taud	2.362	2.220	2.336	2.243	2.198	2.202	2.220	2.197	2.291	2.419	2.464	2.409	(41)	
(42)		Ebn at Noon	844	845	892	875	852	839	837	820	829	833	818	810	(42)	
(43)		Edh at Noon	82	112	112	133	142	142	138	137	117	91	75	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.24	3.50	4.63	5.34	6.05	6.14	5.31	4.32	2.77	1.72	1.18	(44)	
(45)		RadStd	0.13	0.32	0.34	0.42	0.52	0.39	0.30	0.29	0.38	0.29	0.25	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page